

Candidate seat No: _____

Charotar University of Science and Technology [CHARUSAT]
Faculty of Technology and Engineering
Department of Mathematical Sciences
MA243.01 TRANSFORMS AND VECTOR DIFFERENTIAL CALCULUS
1st UNIT TEST

Semester: 3rd Sem. B. Tech. (EC)

Maximum Marks: 30

Date : 13/08/2018 (Monday)

Time: 09:15 am to 10:15 am

Instructions:

- (i) Figures to the right indicate **full** marks.
- (ii) Use of scientific calculator is allowed.
- (iii) Draw figure where it is required.

Q-1 Choose the correct answer from the given options in the following: [06]

1. Which of the following function is not of an exponential order?
(a) $f(t) = e^{t^2}$ (b) $f(t) = e^{at}$
(c) $f(t) = \sin t$ (d) *none of these*
2. Which of the following function does not have Laplace Transform?
(a) $g(t) = t^2$ (b) $g(t) = e^{t^2}$
(c) $g(t) = \frac{\sin t}{t}$ (d) *none of these*
3. If $u_n = \{15, 10, 7, \overset{4}{\uparrow}, 1, -1, 0, 3, 6\}$ then $u_3 =$ _____.
(a) 0 (b) 15
(c) 3 (d) -1
4. If $1 + \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots$ is a geometric series then sum $S =$ _____.
(a) 1 (b) 2
(c) 3 (d) 4
5. Which of the following statement is not true?
(a) $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta$ (b) $\vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \hat{n}$
(c) $\vec{a} + \vec{b} = |\vec{a}| |\vec{b}| \sin \theta$ (d) *none of these*
6. There is no change in magnitude of a vector in $\vec{F}$, if
(a) $\nabla \times \vec{F} \neq 0$ (b) $\nabla \times \vec{F} = 0$
(c) $\text{div}(\vec{F}) \neq 0$ (d) $\text{div}(\vec{F}) = 0$

Q-2 Attempt any Three. [12]

- (a) I. Find Z - transform of sequence $u_n = a^n$.
II. If $L\{f(t)\} = F(s)$ then $L\{e^{at}f(t)\} = F(s - a)$.

(b) I. If $u_n = \{8, 6, 3, -1, 0, 1, 4, 5\}$ then find $Z(u_n)$.

II. Prove that $L\{t^n\} = \frac{n!}{s^{n+1}}, S > 0, n \geq 0$.

(c) Define gradient of a scalar field and Solenoidal vector field. Determine the value of constants "a", so that the following vector is solenoidal, where $\vec{F} = (x + 3y)\hat{i} + (y - 2z)\hat{j} + (x - az)\hat{k}$.

(d) Find the directional derivative of a scalar function $f(x, y, z) = x^2 + xy + z^2$ at the point $A(1, -1, -1)$ in the direction of the line AB, where B is $(3, 2, 1)$.

Q-3 Attempt any Three.

[12]

(a) Find the constants a and b so that surface $ax^2 - zby = (a + 2)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at point $(1, -1, 2)$.

(b) Find Laplace transform of $\frac{\sin 3t}{t}$ and b^t .

(c) Evaluate $L\{t^2 \cos at\}$.

(d) Show that $\vec{V} = yz\hat{i} + xz\hat{j} + xy\hat{k}$ is the irrotational and find a scalar function $u(x, y, z)$ such that $\vec{V} = \text{grad}(u)$.
